Privacy-Preserving Consensus for Pars Network

Pars Network Technical Whitepaper PNP-012

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Abstract

We present Pars Consensus, a privacy-preserving Byzantine fault tolerant (BFT) protocol designed for the Pars Network's dual-layer architecture. Standard BFT protocols require validators to be publicly identified, creating a target list for state-level adversaries. Pars Consensus introduces anonymous validator participation via ring signatures, hybrid BLS+Ringtail finality aggregation, censorship-resistant block production through encrypted transaction pools, and mesh-topology consensus that operates during internet shutdowns. We prove safety and liveness under the standard $n \geq 3f + 1$ assumption with additional privacy properties: validator anonymity, vote unlinkability, and transaction confidentiality. The protocol achieves 2,400 TPS with 1.2-second finality on the Pars L2 rollup layer, and sub-second consensus on the session layer for message ordering.

Keywords: BFT consensus, anonymous validators, ring signatures, censorship resistance, mesh topology, validator privacy, block production

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1 Introduction

Consensus protocols are the foundation of any decentralised network. For the Pars Network, consensus must satisfy constraints beyond those of general-purpose blockchains:

1. **Validator anonymity.** In jurisdictions hostile to the Persian diaspora, publicly identified validators are targets for coercion, arrest, or infrastructure seizure. Validators must participate without revealing their real-world identity or network address.
2. **Censorship resistance.** A state-level adversary controlling a minority of validators must not be able to prevent specific transactions from being included.
3. **Mesh operation.** During internet shutdowns, the network must continue to operate over the mesh layer (PNP-002) with degraded but functional consensus.
4. **Transaction confidentiality.** Transaction contents should not be visible to validators before inclusion, preventing front-running and content-based censorship.

1.1 Relation to Existing Work

Pars Consensus builds on three foundations:

- **Lux Snowman consensus.** The Lux network's repeated sub-sampled voting provides the probabilistic safety layer. Pars adapts the sampling mechanism to preserve voter anonymity.
- **HotStuff BFT.** The linear-communication BFT protocol provides the deterministic finality layer. Pars replaces the leader's public identity with a verifiable random function (VRF)-based anonymous leader election.
- **Ringtail signatures.** Ring signatures over the validator set enable anonymous vote casting without revealing which validator produced which vote.

1.2 Contributions

- A hybrid consensus protocol combining probabilistic safety (Snowman-style) with deterministic finality (HotStuff-style), extended with validator anonymity.
- Anonymous leader election via VRF with ring-signature eligibility proofs.
- Censorship-resistant block production using threshold-encrypted transaction pools (commit-reveal).
- Mesh-topology consensus for operation during internet shutdowns.
- Safety and liveness proofs under the diaspora threat model.

2 System Model

2.1 Network Model

We assume partial synchrony: there exists an unknown Global Stabilisation Time (GST) after which all messages are delivered within a known bound Δ.

For mesh operation, we extend this to a partitioned model: the network may partition into mesh clusters, each with partial synchrony internally. Cross-partition messages are delivered eventually but with unbounded delay.

2.2 Adversary Model

Definition 2.1 (Pars Adversary). *The adversary $\mathcal{A}$ controls at most $f < n/3$ validators (Byzantine) and additionally:*

(a) Can observe all network traffic (global passive adversary).

(b) Can perform traffic analysis on validator communication patterns.

(c) Can target identified validators for coercion.

(d) Cannot break the ring signature or VRF assumptions.

Definition 2.2 (Validator Anonymity). *A consensus protocol provides validator anonymity if $\mathcal{A}$, given the protocol transcript, cannot determine which validator produced which vote with probability better than $1/|V|$ where V is the eligible validator set, except for votes cast by corrupted validators.*

3 Protocol Overview

Pars Consensus operates in three phases per block:

1. **Proposal.** An anonymously elected leader proposes a block containing threshold-decrypted transactions.
2. **Voting.** Validators cast anonymous votes via ring signatures.
3. **Finality.** Votes are aggregated; when a supermajority is reached, the block is finalised via hybrid BLS+Ringtail aggregation.

3.1 Anonymous Leader Election

Definition 3.1 (VRF-Based Leader Election). *Each validator V_i holds a VRF key pair (vk_i, sk_i). For round r, V_i computes:*

$$(\pi_i, y_i) = \mathsf{VRF.Eval}(sk_i, r \| \mathsf{seed}_r)$$

V_i is the leader if $y_i < T$ where T is the target threshold calibrated to elect one leader in expectation.

The leader publishes (π_i, y_i) wrapped in a ring signature over the validator set, proving eligibility without revealing identity.

Theorem 3.1 (Leader Privacy). *Under the ring signature unforgeability and anonymity assumptions, the adversary cannot determine the leader's identity from the election proof, except by being the leader.*

3.2 Block Proposal

The leader constructs the block from the encrypted transaction pool:

1. Users submit transactions encrypted under a threshold public key held by the validator committee (PNP-010).
2. The leader selects transactions by encrypted metadata (gas price commitment, size) without seeing contents.
3. The selected transactions are threshold-decrypted by the committee as part of block execution.

4. Decrypted transactions are executed; invalid ones are discarded.

Theorem 3.2 (Censorship Resistance). *A Byzantine minority controlling $f < n/3$ validators cannot prevent a valid transaction from being included within $O(f)$ rounds, assuming the transaction is submitted to at least one honest validator.*

Proof. Since transaction contents are encrypted until block execution, a Byzantine leader cannot selectively exclude transactions by content. Selection is based on encrypted metadata commitments. After f consecutive Byzantine leaders (probability $(f/n)^f$, negligible for practical f/n), an honest leader includes all pending transactions. □

4 Anonymous Voting

4.1 Ring Signature Votes

Each validator casts a vote as a ring signature over the full validator set.

Definition 4.1 (Ringtail Signature). *A Ringtail signature [3] is a linkable ring signature with sub-linear size. For a ring of n members, the signature size is $O(\sqrt{n})$ rather than $O(n)$.*

Linkability. Votes are linkable *within a round* to prevent double-voting: two votes in the same round by the same validator produce the same linkage tag, enabling detection without deanonymisation.

Definition 4.2 (Vote Unlinkability). *Votes in different rounds are unlinkable: given votes in rounds r and r', the adversary cannot determine whether they were produced by the same validator.*

4.2 Hybrid BLS+Ringtail Finality

For aggregation efficiency, we use a hybrid scheme:

1. **Pre-commit phase.** Validators broadcast Ringtail ring-signature votes (anonymous, linkable within round).
2. **Commit phase.** Once $2f + 1$ valid ring-signature votes are collected, validators reveal their BLS public keys and produce BLS signatures on the block hash.
3. **Aggregation.** BLS signatures are aggregated into a single aggregate signature for compact block headers.

Theorem 4.1 (Finality). *A block is final when $2f + 1$ BLS signatures are aggregated. Under partial synchrony, finality is reached within 3Δ after GST.*

The hybrid approach provides anonymity during voting (ring signatures) and efficiency for finality proofs (BLS aggregation). Validator identities are revealed only *after* the vote outcome is determined, preventing targeted coercion during the voting phase.

5 Censorship-Resistant Block Production

5.1 Encrypted Transaction Pool

Definition 5.1 (Commit-Reveal Transaction Submission). *(i)* ***Commit.*** *User encrypts transaction* tx *under the validator committee's threshold public key:* $c = \mathsf{ThreshEnc}(pk_{\text{comm}}, \mathsf{tx})$. *User broadcasts* (c, meta) *where* meta *contains committed gas price and size.*

(ii) ***Include.*** *Leader includes c in the proposed block based on* meta.

(iii) ***Reveal.*** *After block acceptance, validators cooperatively threshold-decrypt c to obtain* tx. *Invalid transactions are marked failed; fees are still charged.*

5.2 Proposer-Builder Separation

To further resist censorship, Pars separates block building from block proposing:

- **Builders** construct ordered transaction lists from the encrypted pool, competing on fee extraction from metadata.
- **Proposers** (anonymous leaders) select a builder's list and propose it.
- Neither builders nor proposers see transaction contents.

6 Mesh Topology Consensus

6.1 Partition-Tolerant Operation

During internet shutdowns affecting parts of the diaspora, the network partitions into mesh clusters connected via the Pars mesh layer (PNP-002). Each cluster with $\geq 3f_{\text{local}} + 1$ validators runs local consensus.

Definition 6.1 (Mesh Consensus)**.** *In partition mode, each mesh cluster C_k with $|C_k| \geq 4$ validators runs a local BFT instance. Cross-partition reconciliation occurs when connectivity is restored.*

6.2 Reconciliation

When partitions reconnect:

1. Clusters exchange block headers and finality proofs.
2. Conflicting transactions (double-spends across partitions) are resolved by the global validator set via a deterministic conflict resolution rule: the partition with more cumulative stake wins.
3. Non-conflicting transactions from all partitions are merged into the canonical chain.

Theorem 6.1 (Eventual Consistency)**.** *Under the assumption that partitions eventually reconnect, all non-conflicting transactions committed by any partition are eventually included in the global canonical chain.*

6.3 Degraded Mode Guarantees

During partition:

- **Safety.** Each partition maintains local safety ($n_{\text{local}} \geq 3f_{\text{local}} + 1$).
- **Liveness.** Each partition maintains local liveness.
- **Cross-partition safety.** Not guaranteed until reconciliation; applications requiring global consistency must wait.

7 Integration with Pars Subsystems

7.1 Session Layer (PNP-003)

Consensus messages are transmitted over the Pars session layer, which provides:

- End-to-end encryption of consensus messages.
- Onion routing to obscure validator network addresses.
- Forward secrecy for consensus transcripts.

7.2 Shadow Governance (PNP-005, Shadow Governance)

Anonymous governance proposals are ordered by Pars Consensus. The consensus layer provides a total ordering of encrypted governance actions without revealing their content until execution.

7.3 Coercion Resistance (PNP-005)

A coerced validator runs the consensus protocol with their panic credentials, producing valid-looking but ineffective consensus messages that do not contribute to safety violations.

7.4 Lux Network

Pars Consensus inherits the Lux Snowman probabilistic safety mechanism for the initial vote-gathering phase. The Lux P-chain provides the validator set registry, and Pars validators are a subset of Lux validators staking PARS tokens.

8 Safety and Liveness

Theorem 8.1 (Safety)**.** *If $n \geq 3f + 1$, no two honest validators finalise conflicting blocks at the same height.*

Proof. Finality requires $2f + 1$ BLS signatures. Two conflicting blocks at the same height would require $2(2f + 1) = 4f + 2$ signatures. Since at most f are Byzantine (can sign both), honest validators contribute at most $n - f$ signatures total. We need $4f + 2 - f = 3f + 2 > 3f + 1 = n$ honest signatures, a contradiction. □

Theorem 8.2 (Liveness)**.** *After GST, a valid transaction submitted to an honest validator is finalised within $O(f + 1)$ rounds.*

Proof. After GST, messages are delivered within Δ. In each round, the probability of an honest leader is $(n - f)/n > 2/3$. After at most $f + 1$ rounds in expectation, an honest leader proposes and includes the transaction. Honest validators vote within Δ; finality is reached within 3Δ of the proposal. □

Theorem 8.3 (Anonymity)**.** *Under the Ringtail anonymity assumption, the adversary's advantage in determining which honest validator produced a given pre-commit vote is negligible.*

9 Performance

10 Related Work

Lux Snowman. Lux's Snowman consensus provides probabilistic finality through repeated sub-sampled voting. Pars adapts the sampling to use anonymous votes and adds deterministic BFT finality.

Table 1: Pars Consensus benchmarks (100 validators, $f = 33$).

Metric	**Value**
Throughput (TPS)	2,400
Finality latency	1.2 s
Ring signature size	1.8 KB
Aggregate BLS signature	48 bytes
Block header overhead	2.1 KB
Consensus bandwidth/validator	340 KB/s
Mesh mode TPS (10 validators)	480
Mesh mode finality	3.5 s

HotStuff. Yin et al. [1] introduced linear-communication BFT. Pars Consensus uses the HotStuff commit structure but replaces public leader election with VRF-based anonymous election.

Espresso. The Espresso sequencer [2] provides BFT consensus for L2 rollups. Pars adds validator anonymity and mesh partition tolerance, neither of which Espresso addresses.

Ringtail. Ringtail [3] provides sub-linear ring signatures. Pars is the first to deploy Ringtail in a BFT consensus context.

11 Conclusion

We have presented Pars Consensus, a privacy-preserving BFT protocol that provides validator anonymity through ring signatures, censorship resistance through encrypted transaction pools, and mesh-topology operation during internet shutdowns. The hybrid BLS+Ringtail finality mechanism balances anonymity during voting with aggregation efficiency for finality proofs. Safety and liveness are maintained under the standard $n \geq 3f + 1$ assumption, with additional anonymity and censorship-resistance properties critical for diaspora communities operating under authoritarian threat models.

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